Sahaj Bhadja

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Summary

Software engineer with 2+ years building **full-stack AI-enabled products**, **data-intensive workflows**, and **production-grade APIs** with a strong bias for shipping, iteration speed, and measurable business impact.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **full-stack data workflows** used by 500+ users, reducing processing latency from **1.2s to 350ms** through backend optimization and query tuning.
- Implemented **automated validation and audit checks** for 5K+ monthly runs, improving output accuracy and reliability for operations-critical decisions.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **user-facing product features** handling 10K+ daily interactions, translating product requirements into scalable APIs and responsive web interfaces.
- Improved **release velocity** using CI/CD checks and structured defect triage, reducing issue turnaround by **60%** in iterative sprint cycles.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **ML data pipelines** over 100K+ trajectory records, enabling reproducible evaluation workflows and high-confidence model behavior analysis.
- Produced **technical documentation and test scenarios** pre-release, improving handoff quality and reducing implementation ambiguity across teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **AWS-backed API and ETL services** with Glue and S3, improving response times by **30%** under production workloads.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito, improving pipeline confidence by **68%** and reducing regressions.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed **MERN + WebSocket modules** for 2M+ synchronized records across 8 workflows, improving responsiveness of event-driven operational interfaces.
- Automated **Docker-based build and test flows** with Jenkins, reducing release time by **50%** and improving deployment consistency.

Skills

Languages: **Python**, **TypeScript**, JavaScript, SQL, Java, C++, Bash
Full-Stack: React, REST APIs, backend service design, schema modeling, event-driven workflows
AI-Native Workflows: LLM-assisted prototyping, prompt iteration, evaluation-oriented development, automation-first coding
Data & Cloud: PostgreSQL/MySQL/SQL Server, AWS (S3, Glue), containerized deployment, pipeline optimization
Tooling: Git/GitHub, Jenkins, Docker, CI/CD, Agile collaboration, technical documentation

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized REST data-processing API** on AWS CodePipeline and ECS/Fargate with automated CI/CD and validation checks, improving production reliability.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built a **schema-aware NL-to-SQL assistant** across 10+ datasets with validation and explainable outputs, reducing analyst turnaround for ad-hoc data queries.

Freight Audit Agent Prototype | *Python, FastAPI, PostgreSQL, OpenAI API*
- Implemented an **LLM-assisted document extraction workflow** for invoice/PDF normalization with rule-based reconciliation, improving structured field capture and audit explainability.